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Hydrogem Atom Abstraction

Copper-Catalyzed Radical Allene C(sp^2)–H Cyanation

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Dedicated to Professor Kendall N. Houk on the occasion of his 80th birthday

Abstract: While the hydrogen atom abstraction (HAA) from C(sp^3)–H bond has been well explored, the radical-mediated chemo- and regio-selective functionalization of allenic C(sp^2)–H bond via direct HAA from C(sp^2)–H bond of allene remains an unsolved challenge in synthetic chemistry. This is primarily due to inherent challenges with addition of radical intermediates to allenes, regioselectivity of HAA process, instability of allenyl radical toward propargyl radical et al. Herein, we report a copper catalyzed allenic C(sp^2)–H cyanation of an array of tri- and di-substituted allenes with exceptional site-selectivity, while mono-substituted allene was successfully cyanated, albeit with a low yield. In the developed strategy, steric *N*-fluoro-*N*-alkylsulfonamide, serving as precursor of hydrogen atom abstractor, plays a crucial role in achieving the desired regioselectivity and avoiding addition of *N*-centered radical to allene.

Direct insertion of a functional group into a molecule via C–H bond functionalization represents one of most investigated approaches in organic synthetic chemistry. Because of the ubiquity of C–H bonds in organic molecules, by providing access to the desired product with improved atom and step economy over classical synthetic methods, C–H bond functionalization is a particularly powerful toolbox in terms of synthetic efficiency especially when applied to medicinal chemistry, synthesis of natural products and bioactive compounds.[1–3] To address this challenging task, one of the most appealing concepts, which involves the homolytic cleavage of the C–H bond via hydrogen atom abstraction (HAA) procedure, has been successfully devised.[4–10] Tremendous progress has been made for aliphatic C(sp^3)–H bond functionalization via radical HAA process. Nevertheless, allenes are notably absent on the list of suitable substrates as hydrogen donors that have been reported (Figure 1A).[11] In radical reactions, allenes are especially reactive due to unique twisted orthogonal π system that is prone to undergo radical addition to C=C bond.[12] Additionally, to realize allene C–H functionalization, several challenging issues should be addressed: (1) Given that both allenyl and alkenyl C–H bond are sp^2 hybrid, is allenyl C–H bond similar to alkenyl C–H bond, which has so high bond dissociation energy (BDE) that the cleavage of allenyl C–H bond is not easy to access?[13] (2) In the presence of both allenyl C–H bond and other reactive C(sp^3)–H bonds, how could we control the site-selectivity during HAA process? (3) Due to the resonance between

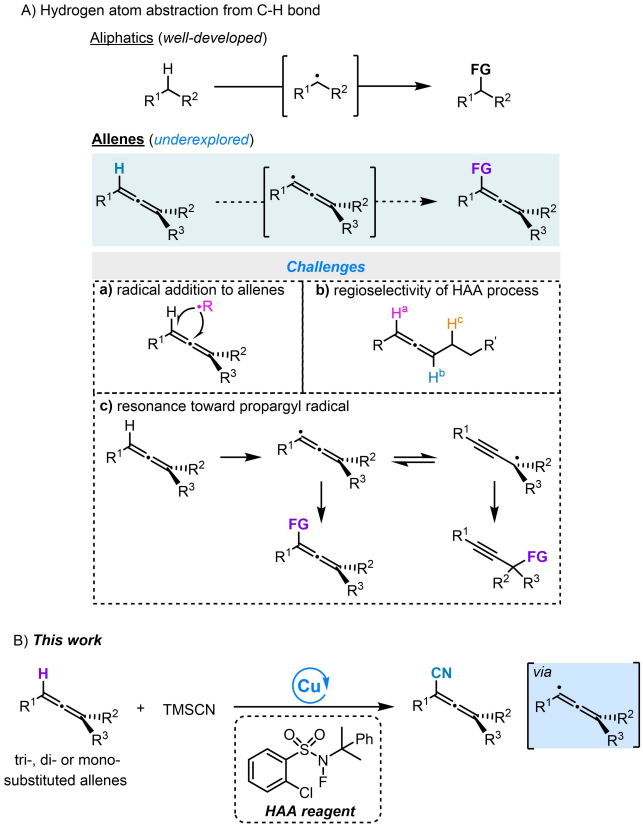


Figure 1. A) Hydrogen atom abstraction from C–H bond. B) This work.

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allenyl radical and propargyl radical,[14] how could we rise to the challenge of selectively trapping allenyl radical? Accordingly, direct radical C–H functionalization of allenes is unknown, although one example of radical allene functionalization via radical addition followed by β-hydride elimination has been reported by Zhang and co-workers.[15]

Some formal functionalizations of allene C–H bond have been reported via sequential C=C addition/C–H cleavage or concerted C=C activation/C–H deprotonation followed by S_E2'.[16–20] They are mechanistically distinct from catalytic allene C–H activation which has been elegantly disclosed by Carreira et al.[21] (Figure 2A) Hou and co-workers also reported rare-earth-catalyzed allene C–H activation to synthesize cyclopropane and cyclopropenes (Figure 2B).[22] Additionally, radical transformations for allene synthesis have been elegantly realized through radical 1,4-addition to 1,3-enynes,[23] transformations from propargylic compounds,[24] 1,2-elimination[25] et al. Despite of these processes, direct C–H functionalization of easily available allenes without the requirement of direction-group and with broader substrate scope in atom- and step-economic mode is very attractive and highly desired. To the best of our knowledge, no related works have been reported to date. Stimulated by the enormous progress of radical HAA chemistry and copper-catalyzed system using *N*-fluoroarylsulfonamides elegantly developed by the Liu group,[26] we hypothesized that allenyl radical could be produced through a hydrogen atom abstraction process mediated by bulky radical species that hardly adds to allene owing to steric hinderance.[27] By utilizing copper salt as catalyst and *N*-fluoro-*N*-alkylsulfonamide as HAA reagent, direct cyanation of allene C–H bonds under mild conditions were achieved with good yields and exclusive selectivities (Figure 1B).

Trisubstituted allene (**1a**) with an allenyl C–H bond and two formal allyl C–H bonds that might compete in HAA process, was subjected to screening of the reaction parameters. Mechanistically, the interaction of a Cu salt/ligand with an HAA reagent, *N*-fluoro-*N*-alkylamide, generated a *N*-centered radical that could abstract hydrogen atom from allene C–H bond to give allenyl radical. Interaction between allenyl radical and LCuCN species would afford cyanated product (**2a**) via reductive elimination of Cu(III) complex. Based on the assumption that the HAA reagent should be capable of regioselective HAA and could also avoid addition to allene, initial experiment was started from screening HAA reagents with the use of CuOAc as catalyst and 4,4′-di-tert-butyl-2,2′-dipyridyl (dtbbpy, **L1**) as ligand. Although CF_3-substituted benzamide failed to provide corresponding product (**A1**, Table 1, entry 1), the reaction using sulfonylamide (**A2**) proceeded smoothly to give desired product (**2a**) with moderate yield and perfect regioselectivity (Table 1, entry 2). Notably, no allyl C–H cyanated product was found. To increase the steric hindrance of *N*-centered radical to avoid its addition to allene, *ortho*-trifluoro sulfonylamide (**A3**) was synthesized and subjected to this condition (Table 1, entry 3). However, a decreasing yield was provided. Instead, *ortho*-chloro sulfonylamide (**A5**) gave the best yield (Table 1, entry 5). Either

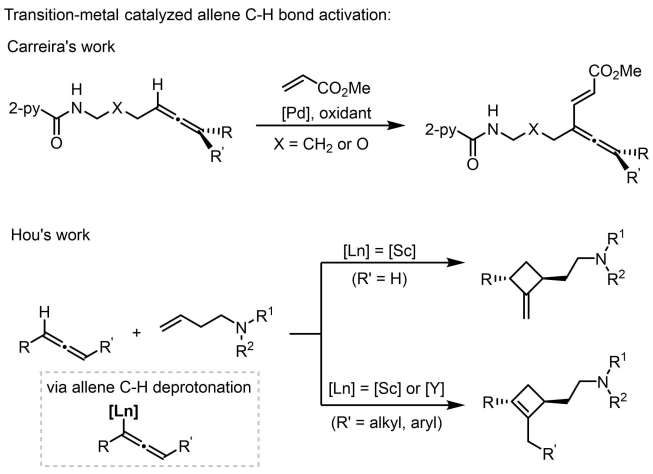
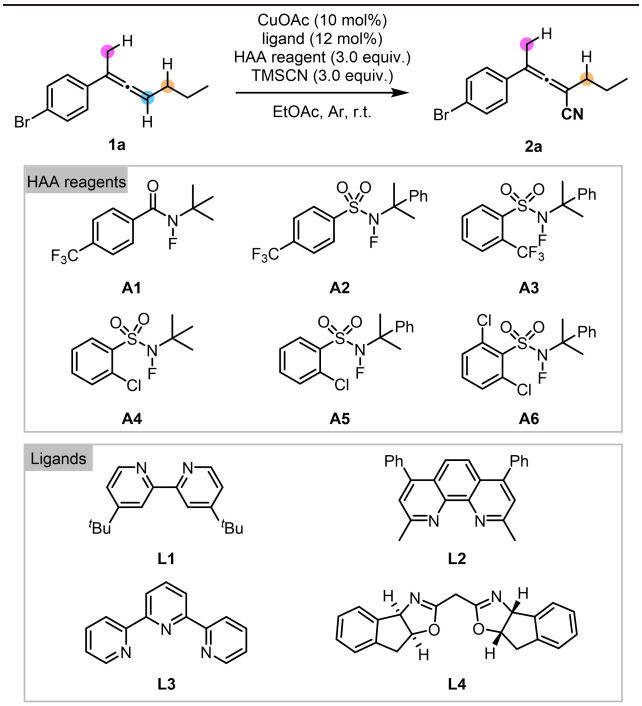


Figure 2. Transition-metal catalyzed allene C–H bond activation.

Table 1: Optimization of reaction conditions.



Entry	HAA reagent	Ligand	Yield [%]
1	**A1**	**L1**	Trace
2	**A2**	**L1**	59
3	**A3**	**L1**	43
4	**A4**	**L1**	21
5	**A5**	**L1**	76
6	**A6**	**L1**	58
7	**A5**	**L2**	9
8	**A5**	**L3**	0
9	**A5**	**L4**	20

Reaction conditions: **1** (0.2 mmol), CuOAc (0.02 mmol), ligand (0.024 mmol), HAA reagent (0.6 mmol), TMSCN (0.6 mmol) in EtOAc (2 mL) under an Ar atmosphere, at room temperature. Yields of isolated products are given.

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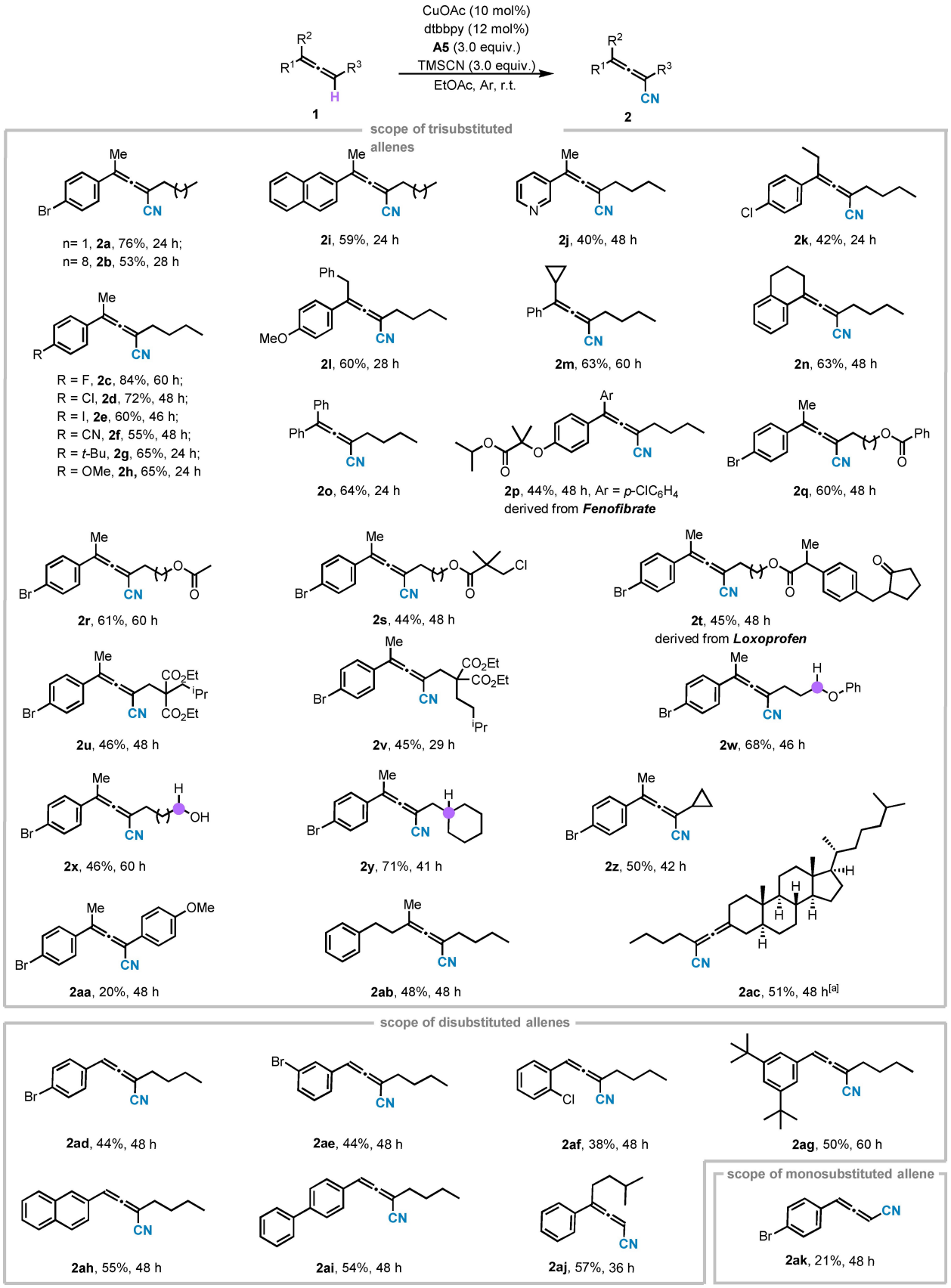


Figure 3. Scope of allenes. Reaction conditions: **1** (0.2 mmol), CuOAc (0.02 mmol), dtbbpy (0.024 mmol), **A5** (0.6 mmol), TMSCN (0.6 mmol) in EtOAc (2 mL) under an Ar atmosphere, at room temperature. Yields of isolated products are given. [a] d.r. = 2.2 : 1.

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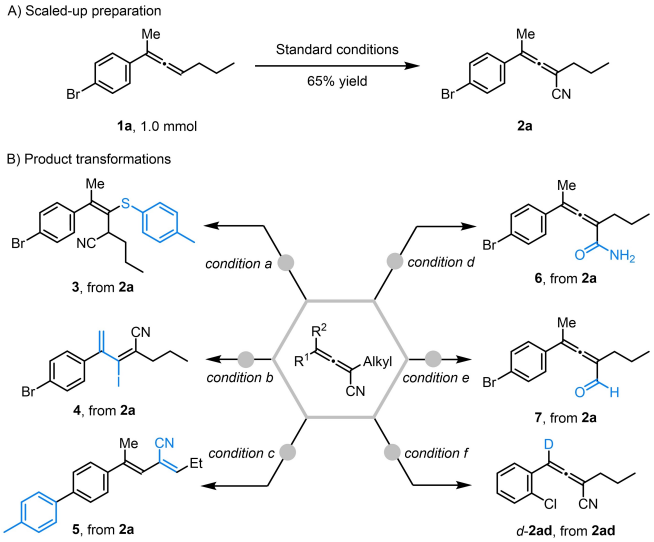


Figure 4. Scaled-up preparation and product transformations. Condition a: **2a** (0.1 mmol), 4-methylbenzenethiol (1.2 equiv.), Et_3N (2.0 equiv.), $CHCl_3$, N_2, rt, 1 d, 54% yield. Condition b: **2a** (0.1 mmol), NIS (*N*-iodosuccinimide, 1.5 equiv.), MeCN, N_2, 80 °C, 1 d, 73% yield. Condition c: **2a** (0.1 mmol), 4-tolylboronic acid (1.5 equiv.), $Pd(OAc)_2$ (1 mol%), PCy_3 (2 mol%), K_3PO_4 (3.0 equiv.), toluene/H_2O = 10:1, N_2, 80 °C, 1 d, 65% yield. Condition d: **2a** (0.1 mmol), NaOH (20 mol%), Na_2CO_3 (1.0 equiv.), H_2O_2 (30% aqueous solution, 4.0 equiv.), EtOH/H_2O = 5:1, N_2, rt, 1 d, 38% yield. Condition e: **2a** (0.1 mmol), DIBAL–H (4.0 equiv.), DCM, N_2, −78 °C, 2 h, 57% yield. Condition f: **2ad** (0.1 mmol), K_2CO_3 (0.3 mmol), $^nBu_4NBr$ (0.1 mmol), toluene (0.5 mL) and D_2O (0.5 mL), N_2, 54% yield, 99% D.

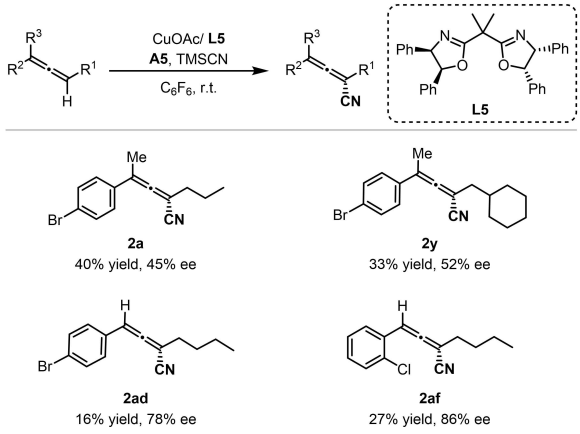


Figure 5. Enantioselective C(sp^2)–H cyanation of allenes.

decreasing or increasing steric hindrance was detrimental to the reaction outcome (Table 1, entries 4 and 6). The use of other ligands such as bathocuproine (**L2**), 2,2′:6′,2″-terpyridine (**L3**) and bis((3aR,8aS)-8,8a-dihydro-3aH-indeno[1,2-d]oxazol-2-yl)methane (**L4**) resulted in poor yields (Table 1, entries 7–9).

The optimized reaction condition for regioselective allene C–H cyanation was used in an investigation of the generality of our protocol. First, the scope of trisubstituted allene was examined (Figure 3). Allene (**1b**) with a long

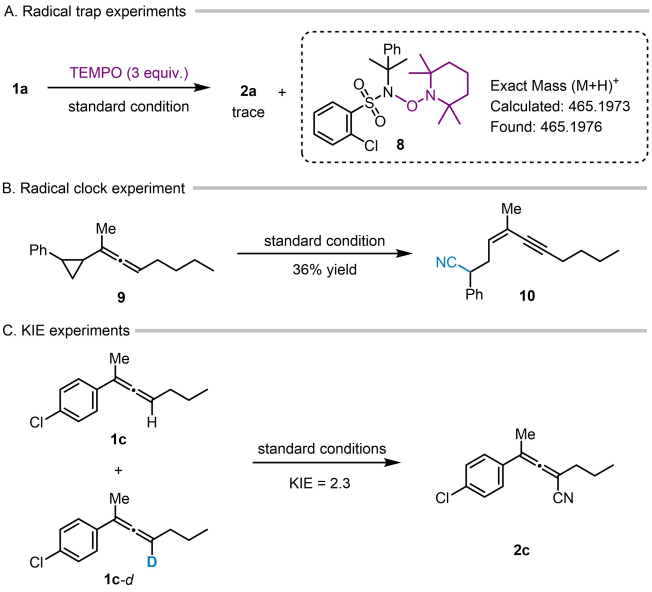
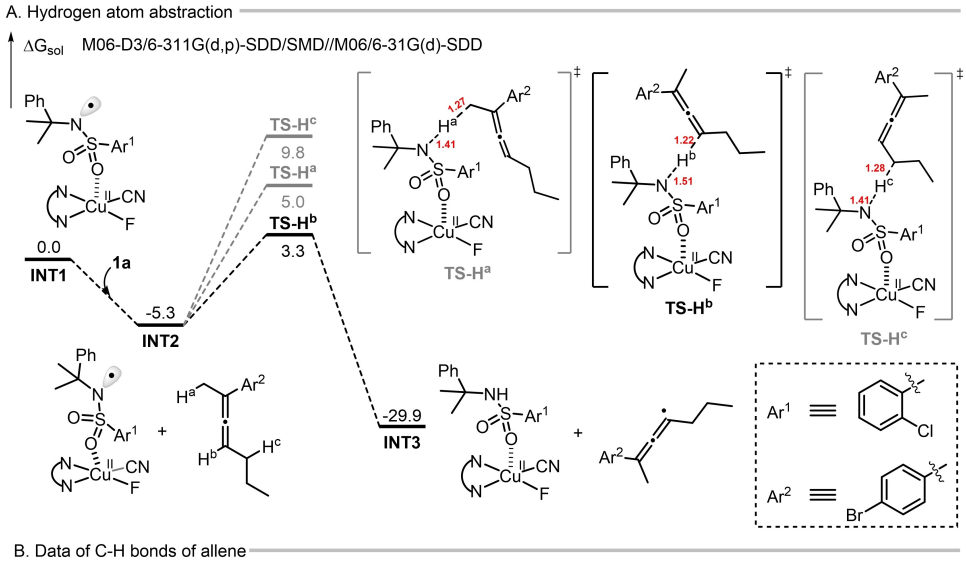


Figure 6. Mechanistic experiments.

pendent aliphatic chain was also suitable for the reaction to provide corresponding cyanated product (**2b**). A portfolio of allenes bearing either electron-donating or -withdrawing aryls were found to be suitable substrates, leading to the products (**2c**–**2h**) in moderate to excellent yields and good regioselectivities. In addition to the phenyl ring, the naphthyl and pyridyl allenes were also well-tolerated, resulting in the formation of corresponding products (**2i**–**2j**) in synthetically useful yields. In lieu of methyl group, allenes with the ethyl, benzyl, cyclopropyl and tetrahydronaphthyl group were readily converted to desired cyanated compounds (**2k**–**2n**). It is noteworthy that the presence of benzylic C–H bonds (**2l**) with low BDE had no effect on the site-selective hydrogen abstraction from allenic C–H bond. In terms of the example of **2m**, "radical clock" process-driven ring-opening product derived from propargylic radical, a resonance structure of allenic radical, was not detected. 1,1-Biaryl allenes were also compatible to yield the desired products (**2o** and **2p**). Remarkably, successful conversion of **1p** derived from Fenofibrate for the treatment of hyperlipidemia, showcased the potential for application of this protocol in late-stage elaboration of bioactive compounds or drugs. Many structural units, such as ester (**2q**–**2v**), chloride (**2s**), and ketone (**2t**) derived from Loxoprofen, were well tolerated in this protocol. Especially, some reactive C(sp^3)–H bonds, including phenyl ether (**2w**), unprotected alcohol (**2x**), and cyclic tertiary C(sp^3)–H bond (**2y**) were intact under this condition. Of note, allenic C–H bond (**1z**) adjacent to cyclopropyl ring was readily cyanated and no ring-opening product via "radical clock" process was formed. The reaction with 1,3-diaryl allene gave desired product (**2aa**) successfully, albeit with a decreasing yield probably attributing to high steric hinderance. In addition to aryl-substituted allenes, allene substrates with alkyl groups or complex molecular skeleton, were subjected to our

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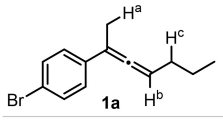
C-H bonds of allene		BDE (kcal/mol)	ΔG‡(kcal/mol)
	C-H^a	88.7	10.3
	C-H^b	77.6	8.6
	C-H^c	83.7	15.1

Figure 7. DFT calculations. A) Hydrogen atom abstraction. B) Data of C–H bonds of allene.

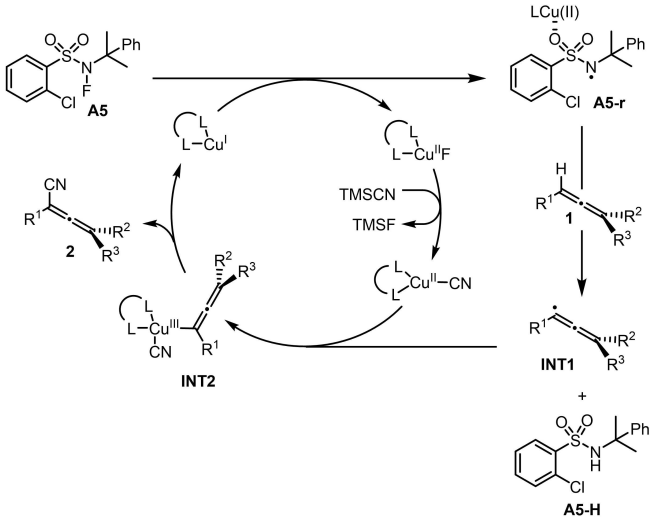


Figure 8. Proposed mechanism.

protocol, giving rise to desired allenyl nitriles (**2ab** and **2ac**) with excellent regioselectivities and moderate yields.

The compatibility of di- and mono-substituted allenes with two or three allenic C–H bonds that might compete in HAA process (Figure 3, bottom) were explored next. Disubstituted allenes pedant with ortho-, meta-, and para-halogenated aryl group smoothly underwent site-specific HAA from allenyl C–H bonds near aliphatic chain with less steric hinderance, leading to a set of trisubstituted allenes (**2ad**–**2ag**) with synthetically useful yields and excellent regioselectivities. Naphthyl and biphenyl allenes (**1ah**–**1ai**) were found to be suitable substrates giving rise to the desired products. In the case of 1,1-arylalkyl allene, the abstraction of allenic C–H bond also proceeded readily to furnish cyanated allene (**2aj**). Interestingly, when using monosubstituted allene (**1ai**), the exclusive regioselectivity of the HAA process occurred at terminal C–H bond instead of benzyl C–H bond. However, a decreasing yield was given, probably owing to addition of *N*-centered radical from HAA reagent (**A5**) to allene leading to byproduct.

The scaled-up preparation of **1a** proceeded readily and delivered a comparable yield, showing the practicality of the protocol (Figure 4A). The products can be conveniently transformed into other valuable compounds (Figure 4B). For example, the allene in **2a** was converted to a tetra-substituted alkene (**3**) by nucleophilic addition with *p*-toluenethiol. Treating the allene with *N*-iodosuccinimide (NIS) generated iodo-1,3-diene (**4**) through C–H bond elimination. Similar elimination occurred when **2a** was then subjected to the Suzuki coupling with 4-tolylboronic acid to afford biaryldiene (**5**). The cyano group in **2a** could be substituted with transformable functional units, such as amide (**6**) or aldehyde (**7**), which offer a platform for further elaborations. Deuteration of allenyl C–H of **2ad** with D_2O in presence of K_2CO_3 and *n*-Bu_4NBr readily gave rise to *d*-**2ad** in 99 % D-incorporation.

We also examined the catalytic enantioselective cyanation of allenyl C–H bond (Figure 5). The reactions with di- and tri-substituted allenes (**1a**, **1y**, **1ad** and **1af**) gave

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moderate to good enantioselectivities, while the yields need further improvement.

Radical trapping experiment was carried out to probe the radical species involved in the reaction (Figure 6A). In the presence of TEMPO, the allene C–H cyanation was inhibited and *N*-centered radical generated in situ was successfully captured by TEMPO to give **8** that was detected by HRMS. Furthermore, a radical-clock experiment was designed to detect allenyl radical (Figure 6B), and desired ring-opening product (**10**) was afforded. These results confirmed the formation of *N*-centered radical and allenyl radical in the transformation. The measured product isotope effect of 2.3 for the KIE experiment of reactions with **1c** and **1c–d** suggests that the intermolecular HAA process might be the rate-limiting step in this process (Figure 6C).

To obtain a deeper understanding of chemo- and regioselectivity of the cyanation of allenyl C(sp^2)–H bond, density functional theory (DFT) calculations were carried out (Figure 7). For the example of **1a** with an allenyl C(sp^2)–H bond and two allylic C(sp^3)–H bonds, the activation energy ($\Delta G^{\neq}$) for HAA process involving these C–H bonds and the ΔG of this step were calculated and shown in Figure 7A, demonstrating that HAA from allenyl C(sp^2)–H bond requires the lowest activation energy (8.6 kcal/mol) and is exergonic by −29.9 kcal/mol. In addition, to explore the regioselectivities of these reactions, BDEs of C–H bonds of substrate **1a**, were calculated and shown in Figure 7B. These results illustrate that allenyl C(sp^2)–H bonds with lower BDE, are prone to be readily cyanated through HAA process with relatively lower activation barrier and more exergonic energy. Of note, BDE of allenyl C(sp^2)–H^b bond is even lower than that of some activated C(sp^3)–H bonds, such as toluene (89.7 kcal/mol), allylic C–H bond (about 88 kcal/mol), ethanol (95.9 kcal/mol), tetrahydrofuran (THF, 92.1 kcal/mol) et al. This also explains exclusive selectivity at the allenyl C(sp^2)–H bond even in the presence of some activated C(sp^3)–H bonds.

Based on this study of the mechanism, a plausible pathway is outlined in Figure 8. The interaction between Cu(I) and the HAA reagent (**A5**) gives rise to a Cu(II)-bound *N*-centered radical (**A5-r**). This radical (**A5-r**) abstracts selectively a hydrogen atom from allene C–H bond to provide the allenyl radical (**INT1**). The interaction between Cu(II)F complex and TMSCN delivers a Cu(II)CN species, which upon interception by **INT1** furnishes **INT2**. The ensuing reductive elimination of **INT2** furnishes the final product (**2**) and regenerates Cu(I) to perpetuate the catalytic cycle.

In summary, a novel copper-catalyzed site-selective radical allenyl C(sp^2)–H functionalization has been described. The key to the success of this reaction lies in the regioselective intermolecular HAA process mediated by a bulky HAA reagent, *N*-fluoro-*N*-alkylsulfonylamide, thus avoiding its addition to allene. A set of valuable tetra-, tri- and di-functionalized allene derivatives have been obtained in synthetically useful yields in this way. DFT calculations provide theoretical evidence supporting the regioselectivity of this HAA process. The method features broad functional group compatibility and high product diversity, and allows a much broader chemical space for radical C(sp^2)–H functionalization.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: C–H Functionalization · Copper Catalysis · Cyanation · Radical Reaction · Regioselectivity

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